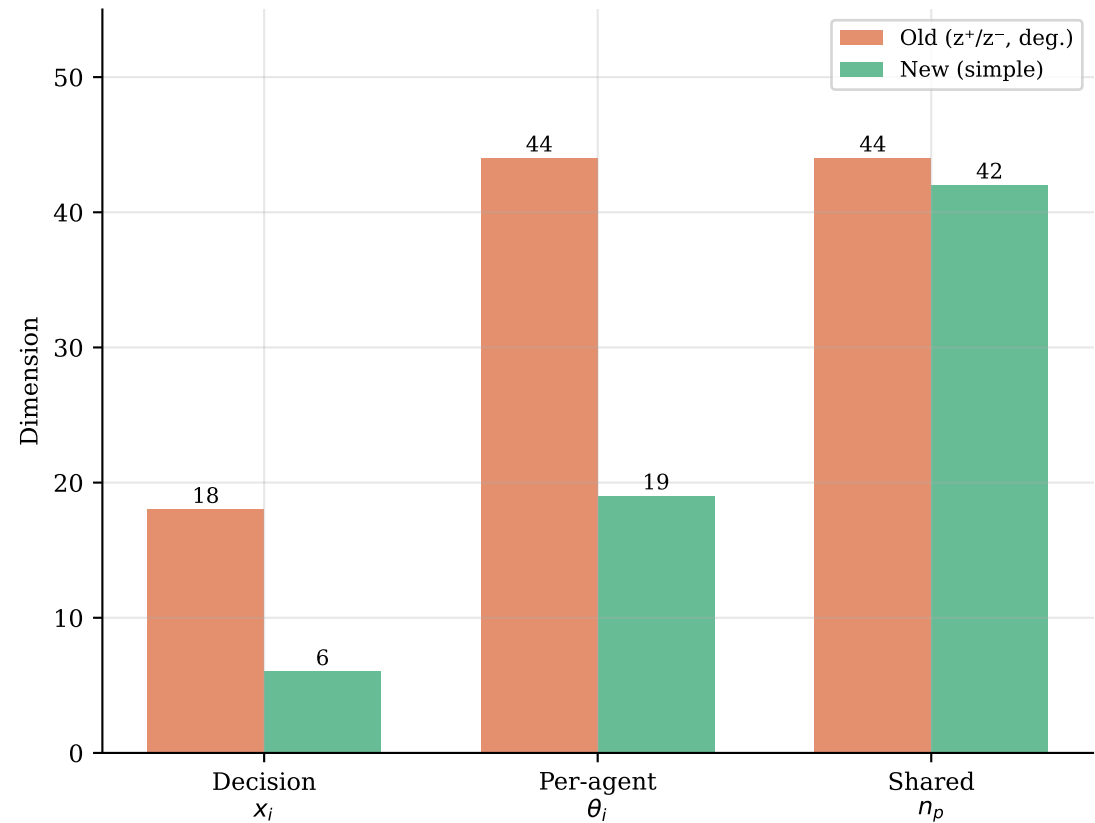


Simple Game: Problem Structure vs. Old Formulation

Old Formulation



Formulation Comparison

Aspect	Old formulation	New (simple)
Decision x_i	$[p, z+, z-]$ in $\mathbb{R}^{18}$	p in $\mathbb{R}^6$
Cost Hessian Q_i	block-diag + deg.	γI_6 (diagonal)
Imbalance penalty	linear $\gamma+z++\gamma-z-$	$(1/2)*\gamma* p-pDA ^2$
Ramp constraints	yes	no
Degradation term	$a_{deg} * p^2$	no
z-update dimension	$2H = 12$ (one-sided)	$H = 6$ (bilateral clip)
Per-agent θ_i	$n_p = 44$	$3H+1 = 19$
mpQP param total	62	37